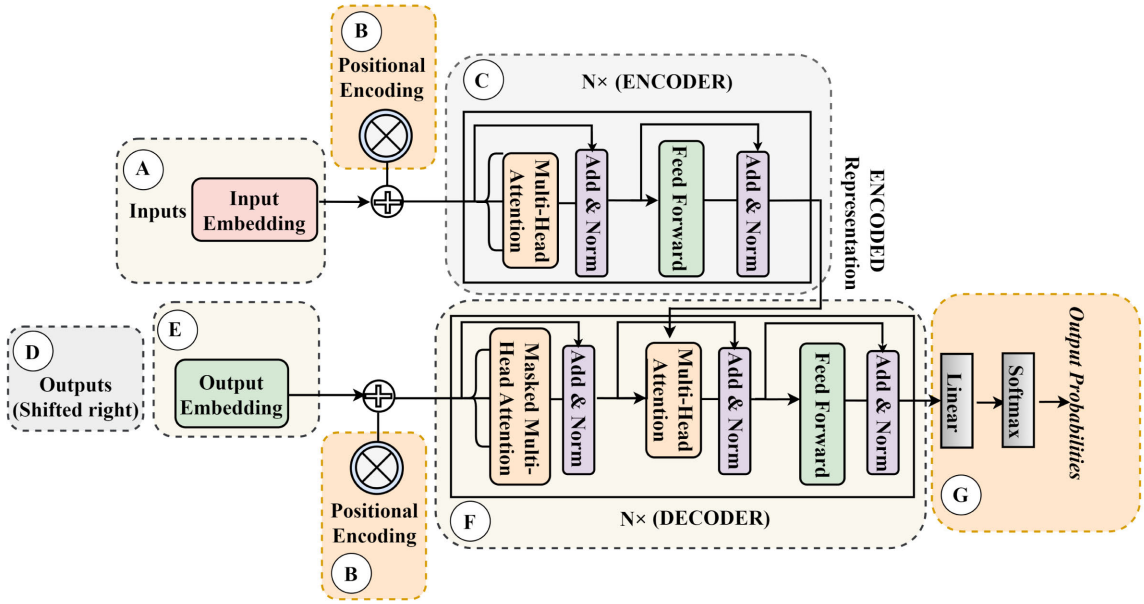


FIGURE 6. Architecture of a Transformer model.

To address the problem, positional encoding is used to encode the position of each word in the input sequence as a collection of integers. The transformer model uses integer, input embedding and positional encoding to help GPT in understanding sentence word order and provide grammatically accurate and semantically appropriate output [60]. The positional encoding part is shown in Figure 6B.

- **Encoder** The encoder is a crucial component of the neural network which is responsible for processing the input text. Its primary function is to generate a series of hidden states that represent the input text in a meaningful way [61]. Then, it uses a series of self-attention layers that are often referred to metaphorically as "voodoo magic," emphasizing their complex and powerful ability to capture relationships between different elements in the input text. In the transformer, the encoder is used in more than one layer. This section is depicted in Figure 6C comprehensively.
- **Outputs (shifted right)** During the training process, the decoder in the transformer model learns to predict the next word in a sequence by analyzing the preceding words. This is achieved through a mechanism known as autoregressive training. The decoder's ability to predict the next word is critical for generating coherent and contextually relevant sequences. Additionally, the GPT (GPT-3) is also trained on a massive amount of text data, that helps it to generate sense while writing any content. Besides, several corpus including the Common Crawl web corpus, the BooksCorpus dataset, and the English Wikipedia are also used during the common issue. Figure 6D highlights the transformer's outputs (shifted right) module.
- **Output Embeddings**
  Input embeddings, which contain text are not directly recognized by the model. Therefore, the output must be converted to a format known as "output embedding." Similar to input embeddings, output embeddings undergo positional encoding, enabling the model to understand the order of words in a sentence [62]. In machine learning, the loss function evaluates the difference between a model's prediction and the objective value. Loss functions are essential for complex GPT language models. The loss function modifies a portion of the model to increase accuracy by reducing the discrepancy between predictions and targets. The change improves the overall performance of the model. The loss function is calculated during training, and the model parameters are modified. In the inference process, the output text is created by mapping the predicted probability of each token in the model to the corresponding token in the vocabulary. The output embedding part is illustrated in Figure 6E.
- **Decoder**
  The decoder processes both positionally encoded input and output embeddings. Positional encoding is crucial for the model to understand the sequential order of the tokens in both the input and output sequences. The positional information helps the decoder effectively capture the structure within the sequences. The decoder has an attention mechanism that helps to improve the output's quality by leveraging contextual information received from the encoder. The primary function of the decoder is to create output sequences based on the encoded input sequences. It generates a sequence of tokens, often representing words or